

Chapter 3: Kinematics

3.1 Displacement and Velocity

Displacement is a vector quantity describing the change in position of an object. Average velocity is the displacement divided by the time interval over which that displacement occurs.

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3.2 Projectile Motion

A projectile launched with initial speed u at angle θ above the horizontal can be analyzed by decomposing its motion into independent horizontal and vertical components. The horizontal component of velocity is $u \cos(\theta)$, and it remains constant throughout the flight because there is no horizontal acceleration when air resistance is neglected.

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3.3 Range of a Projectile

The range R of a projectile launched and landing at the same height is given by $R = (u^2 \sin(2 \theta)) / g$, where g is the acceleration due to gravity. The maximum range occurs when the launch angle θ equals 45 degrees.

Chapter 4: Dynamics

4.1 Newtons Second Law

Newtons second law states that the net force acting on an object equals the rate of change of its momentum. For constant mass this reduces to $F = m * a$, where a is the acceleration produced by the net force F .